Aryan Mondal

aryanmondal@umass.edu | LinkedIn | GitHub | Scholar

Education

University of Massachusetts Amherst — Amherst, MA
MS in Computer Science — *Sep 2026 – May 2028*

- **Coursework**: COMPSCI 660 Advanced Information Assurance, COMPSCI 565 Advanced Digital Forensic Systems, COMPSCI 589 Machine Learning

SRM Institute of Science and Technology — Chennai, India
B.Tech in Computer Science and Engineering, CGPA: 8.39/10 — *Sep 2022 – May 2026*

Experience

Security Intern — Feb 2025 – Mar 2025
Thirumoolar IT Solutions — *Chennai, Tamil Nadu*

- Assessed 3 internal web applications with Burp Suite, OWASP ZAP, and Nmap, then reviewed their source code by hand to confirm which issues were actually exploitable.
- Reported 40+ findings, 4 rated critical, including SQL injection and XSS, each with reproduction steps and a recommended fix.

Research

A Robust and Lightweight Authentication Protocol for UAV Swarm Communications using HKDF-based Key Derivation and AEAD — Jul 2026
IEEE CONECCT 2026

- Redesigned the security layer of a published UAV protocol, replacing its chaotic-map keystream with ECDSA-signed ephemeral ECDH (forward secrecy), HKDF per-packet keys, and ChaCha20-Poly1305, chosen over software AES-GCM to avoid cache-timing leaks on chips without AES hardware.
- Derived a fresh key for every packet from the UAV ID and a monotonic counter, and authenticated the ID, counter, and timestamp as associated data, so tampered packets fail the MAC check and replays are rejected by a timestamp window and counter check.
- Simulated swarms of 10–100 UAVs in *UavNetSim* (Python/SimPy): delivery stayed above 99.8% and star-topology delay rose only from 5 to 15 ms, at 32 bytes of overhead and an estimated 150–250 μs of crypto per packet on an ARM Cortex-M4.

Lightweight Hybrid DNN-GNN Architecture for Network Intrusion Detection with Adaptive Late Fusion — Apr 2026
ICNWC 2026

- Built an ∼80K-parameter PyTorch intrusion detector that fuses a 4-layer DNN and a shallower 8K-parameter MLP through a learned weight, reaching 99.64% accuracy and a 0.08% false positive rate on 9,000 held-out CIC-IDS-2017 flows, on par with published 28M–100M-parameter models.
- Ran ablations showing the 8K-parameter branch alone matched the full hybrid (99.64%) and outperformed the 68K-parameter DNN (99.52%), consistent with the learned fusion weight shifting toward the MLP branch from 31% to 61% during training.

Quantum Key Distribution for SCADA in Power Grids: Security Architectures and Practical Applications — Nov 2025
ICETCE 2025

- Surveyed over 30 works on quantum key distribution (QKD) for SCADA and cyber-physical systems, including field deployments in Tokyo, Vienna, and China; mapped QKD architectures to grid threats such as false data injection and man-in-the-middle attacks; and compared four anomaly classifiers on simulated data.

Technical Skills

Programming Languages: Python, C, C++, Bash, SQL
Security: Web application testing (SQLi, XSS), manual code review, applied cryptography (ECDH, HKDF, AEAD), protocol design, packet analysis, ML-based intrusion detection
ML: PyTorch, scikit-learn, Bayesian optimization
Tools: Claude Code, Burp Suite, OWASP ZAP, Nmap, Wireshark, Scapy, NetSquid, Linux, Git